Stochastic Finance and Risk Modelling

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5 - Multi-period model of financial market

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Outline

Discrete-Time Market Model

Arbitrages and Martingale Measures

Payoffs and Derivatives

Complete Market

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Multi-periode model

- Consider a market consisting of 1 riskless asset and d risky assets that can be traded at times $t = 0, \ldots, T$, where $T \in \mathbb{N}$ is the number of periods.
- We aim to extend the previous one-period model to multi-period model.
- Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space endowed with some filtration $\mathbb{F} = (\mathcal{F}_t)_{t=0,\ldots,T}$.
- As the current prices $S_0^1, \ldots, S_0^d \in \mathbb{R}_+$ are observable, we may assume that $\mathcal{F}_0 = \{\emptyset, \Omega\}$.
- We assume that $S_t^1, \ldots, S_t^d$ are $\mathcal{F}_t$-measurable :

$$\sigma(S_t^1, \ldots, S_t^d) \subset \mathcal{F}_t.$$

An example of explicit construction

- The riskless asset S^0 is given by $S_t^0 := (1+r)^t$, where $r \geq 0$ denotes the interest rate.
- Let $\Omega := \underbrace{\mathbb{R}_+^d \times \cdots \times \mathbb{R}_+^d}_{T \text{ product}}$ be the probability space consisting of elements $\omega = (\omega_1, \ldots, \omega_T)$ with $\omega_t := (\omega_{t,1}, \ldots, \omega_{t,d}) \in \mathbb{R}_+^d$. Let $\mathcal{F} := \mathcal{B}(\Omega)$.
- For $i = 1, \ldots, d$ and $t = 1, \ldots, T$, define the projection function $S_t^i : \Omega \to \mathbb{R}_+$ by $S_t^i(\omega) := \omega_{t,i}$, where S_t^i stands for the price of asset i at time t. Denote $S_t := (S_t^1, \ldots, S_t^d)$, then $S_t(\omega) = \omega_t$.

$$t \mapsto \omega = (\omega_1, \ldots, \omega_T) \mapsto S_t(\omega) = \omega_t.$$

- Set $\mathcal{F}_t := \sigma(S_1, \ldots, S_t)$ for $t = 1, \ldots, T$. One may verify $\mathcal{F}_T = \mathcal{F}$.
- Define the discounted processes by $\tilde{S}$, i.e. $\tilde{S}_t^i = \dfrac{S_t^i}{S_t^0}$, $i = 0, \ldots, d$, $t = 0, \ldots, T$.

Self-financing strategy

Definition (Self-financing strategy)

A strategy is identified by a predictable stochastic process $\left(\xi_t^0, \xi_t := (\xi_t^1, \ldots, \xi_t^d)\right)_{t=1,\ldots,T}$ taking values in $\mathbb{R}^{d+1}$. A strategy is called self-financing if

$$(\xi_t^0, \xi_t) \cdot (S_t^0, S_t) = (\xi_{t+1}^0, \xi_{t+1}) \cdot (S_t^0, S_t), \quad t = 1, \ldots, T-1.$$

Interpretation : Neither new investment is added nor existing capital is withdrawn. The portfolio's rebalancing is achieved at constant value.

Portfolio

Definition (Portfolio)

A portfolio associated to a self-financing strategy $(\xi_0, \xi) = (\xi_t^0, \xi_t)_{t=1,\dots,T}$ is given by an adapted process $(V_t^{\xi_0,\xi})_{t=0,\dots,T}$ as follows :

$$\begin{cases} V_0^{\xi_0,\xi} = (\xi_1^0, \xi_1) \cdot (S_0^0, S_0) = \xi_1^0 + \sum_{i=1}^{d} \xi_1^i S_0^i, \\ V_t^{\xi_0,\xi} = (\xi_t^0, \xi_t) \cdot (S_t^0, S_t) = \xi_t^0 (1+r)^t + \sum_{i=1}^{d} \xi_t^i S_t^i, \quad t = 1, \dots, T. \end{cases}$$

The variation of the portfolio's value only comes from the variation of the processes' price :

$$\begin{aligned} V_{t+1}^{\xi_0,\xi} - V_t^{\xi_0,\xi} &= (\xi_{t+1}^0, \xi_{t+1}) \cdot (S_{t+1}^0, S_{t+1}) - (\xi_t^0, \xi_t) \cdot (S_t^0, S_t) \\ &= (\xi_{t+1}^0, \xi_{t+1}) \cdot ((S_{t+1}^0, S_{t+1}) - (S_t^0, S_t)) \end{aligned}$$

Characterization of self-financing strategies

Proposition

Let $(\xi_t^0, \xi_t)_{t=1,\dots,T}$ *be a trading strategy. The following claims are equivalent :*

(i) $(\xi_t^0, \xi_t)_{t=1,\dots,T}$ *is self-financing ;*

(ii) $(\xi_t^0, \xi_t) \cdot (1, \tilde{S}_t) = (\xi_{t+1}^0, \xi_{t+1}) \cdot (1, \tilde{S}_t), \quad t = 1, \dots, T-1$;

(iii) $\dfrac{V_t^{\xi_0,\xi}}{S_t^0} =: \tilde{V}_t^{\xi_0,\xi} = V_0^{\xi_0,\xi} + \displaystyle\sum_{k=1}^{t} \xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1}), \quad t = 0, \dots, T.$

Proposition

Let $(\xi_t)_{t=1,\dots,T}$ *be an* $\mathbb{F}-$*predictable process taking values in* $\mathbb{R}^d$*. For every* $x \in \mathbb{R}$*, there exists a unique predictable process* $(\xi_t^0)_{t=1,\dots,T}$ *taking values in* $\mathbb{R}$ *s.t. the process* $(\xi_t^0, \xi_t)_{t=1,\dots,T}$ *is a self-financing strategy whose portfolio starts with* $V_0^{\xi_0,\xi} = x$ *and* $V_t^{\xi_0,\xi} = x + \sum_{k=1}^{t} \xi_k \cdot (\tilde{S}_k - \tilde{S}_{k-1})$ *for all* $t = 1, \dots, T$*. In this case, we write* $V^{\xi_0,\xi} = V^{x,\xi}$*.*

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Arbitrage

Definition

A self-financing strategy (ξ_0, ξ) yields an arbitrage opportunity if the corresponding portfolio's value process satisfies

$$V_0^{\xi_0,\xi} = 0, \quad \mathbb{P}[V_T^{\xi_0,\xi} \geq 0] = 1, \quad \mathbb{P}[V_T^{\xi_0,\xi} > 0] > 0.$$

No Arbitrage condition : $V_T^{0,\xi} \geq 0$ $\mathbb{P}$-a.s. for some $\xi \Longrightarrow V_T^{0,\xi} = 0$ $\mathbb{P}$-a.s.

Proposition

There exists an arbitrage opportunity if and only if there exists $t \in \{1, \ldots, T\}$ and an $\mathcal{F}_{t-1}-$measurable random vector ξ_t in $\mathbb{R}^d$ s.t.

$$\begin{cases} \xi_t \cdot (\tilde{S}_t - \tilde{S}_{t-1}) \geq 0 \ \ \mathbb{P}\text{-}a.s., \\ \mathbb{P}\left[\xi_t \cdot (\tilde{S}_t - \tilde{S}_{t-1}) > 0\right] > 0. \end{cases}$$

- The no arbitrage condition must hold at every trading period.

Martingale measure

Definition (Martingale measure)

A probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{F})$ is said to be a martingale measure or risk-neutral measure if the discounted price processes $(\tilde{S}_t^1, \ldots, \tilde{S}_t^d)_{t=0,\ldots,T}$ are martingales under $\mathbb{Q}$, i.e.

$$\mathbb{E}_{\mathbb{Q}}[\tilde{S}_t^i] < \infty \text{ and } \tilde{S}_{t-1}^i = \mathbb{E}_{\mathbb{Q}}[\tilde{S}_t^i | \mathcal{F}_{t-1}], \;\; 1 \leq t \leq T.$$

A martingale measure $\mathbb{Q}$ is called an equivalent martingale measure if it is equivalent to $\mathbb{P}$.

Theorem (FTAP)

The market is arbitrage-free, i.e. there is no arbitrage opportunity, if and only if there exists an equivalent martingale measure $\mathbb{Q}$.

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Payoffs and European derivatives

Definition

A payoff is identified by a $\mathcal{F}_T-$measurable r.v.. An European derivative written on the assets $S^1, \ldots, S^d$ is a contract allowing its holder to receive at T a payoff equal to $F(S^1, \ldots, S^d)$, where F is a measurable function.

- American options will be introduced later.
- Exotic derivative : path-dependent derivatives $F(S^0, \ldots, S^d)$.
- European vanilla derivatives : $F(S_T^0, \ldots, S_T^d)$.

Examples of derivatives I

- European call option on asset i of strike K :

$$H = (S_T^i - K)_+$$

- European put option on asset i of strike K :

$$H = (K - S_T^i)_+$$

- European call option on the index/basket of d actifs :

$$H = \left(\sum_{i=1}^{d} w_i S_T^i - K \right)_+$$

Examples of derivatives II

- Asian call option on asset i of strike K :

$$H = \left(\frac{1}{T} \sum_{t=1}^{T} S_t^i - K \right)_+$$

- Look back call option on asset i of strike K :

$$H = \left(\max_{t=1,\dots,T} S_t^i - K \right)_+$$

- Barrier (up-and-out) call option on asset i of strike K :

$$H = (S_T^i - K)_+ \mathbf{1}_{\{S_t^i < B, t=1,\dots,T\}}$$

Replicable European payoff I

Definition (Replicable payoff)

A European payoff H is said to be replicable if there exists a self-financing strategy $(\xi^0, \xi) = (\xi_t^0, \xi_t)_{t=1,\dots,T}$ s.t.

$$H = V_T^{\xi_0,\xi} = (\xi_T^0, \xi_T) \cdot (S_T^0, S_T) \ \ \mathbb{P}\text{-a.s.}$$

This strategy, or the corresponding portfolio, is called the replication of the payoff H.

Replicable European payoff II

Proposition

Let the market be arbitrage-free and H be some replicable European payoff. Then for any equivalent martingale measure $\mathbb{Q}$ and any replicating strategy, the corresponding discounted portfolio's value must satisfy

$$\tilde{V}_t^{\xi_0,\xi} = \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{S_T^0}\middle|\mathcal{F}_t\right] \ \mathbb{Q}\text{-a.s.}, \quad t = 0, \ldots, T.$$

- In particular, $(\tilde{V}_t^{\xi_0,\xi})_{t=0,\ldots,T}$ is a $\mathbb{F}-$adapted positive martingale under $\mathbb{Q}$.
- $V_0^{\xi_0,\xi} = \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{S_T^0}|\mathcal{F}_0\right] = \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{S_T^0}\right]$ as $\mathcal{F}_0 = \{\emptyset, \Omega\}$.

No-arbitrage price I

Assume that the market is arbitrage-free.

Definition

We call π_0 the arbitrage-free price of an European payoff H if there exists an $\mathbb{F}-$adapted process X s.t.

$$X_0 = \pi_0 \quad \text{and} \quad X_T = H$$

and the new market consisting of $d+2$ assets $(S^0, \ldots, S^d, X)$ is still arbitrage-free.

No-arbitrage price II

Proposition

The set of arbitrage-free prices of any European payoff H is given as

$$\left\{\mathbb{E}^{\mathbb{Q}}\left[\frac{H}{S_T^0}\right] : \mathbb{Q} \text{ equivalent martingale measure s.t. } \mathbb{E}^{\mathbb{Q}}\left[\left|\frac{H}{S_T^0}\right|\right] < \infty\right\}.$$

Proposition

Assume the No Arbitrage condition. Let H be a replicable European derivative. Then H has a unique price $V_0^{\xi_0,\xi} = \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{S_T^0}\right]$, where (ξ_0, ξ) is an arbitrary replicating portfolio and $\mathbb{Q}$ is an arbitrary equivalent martingale measure.

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Complete market

Definition

A market is called complete if all the payoffs are replicable.

Proposition

A no-arbitrage market is complete if and only if the equivalent martingale measure is unique.